From Stock to Robot-Adapted VLM Backbones: A Frozen-Feature Study on ALOHA and LIBERO-Goal

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***Abstract*—We test whether frozen features from a robot-adapted vision–language backbone help a newly trained action policy compared with its stock lineage origin. We read Qwen3-VL-2B and the GR00T N1.7 vision–language subtree at language layer 16 and independently train the same action-head architecture within each benchmark. Across three training seeds, GR00T achieves 52.2% mean success on ALOHA Transfer Cube, compared with Qwen's 28.7%, and leads in every seed pair. On two-camera LIBERO-Goal, both average 89.8%, with mixed seed-level differences. Behavioral analysis locates the ALOHA advantage in completion after an intermediate contact stage; Qwen also reaches that stage later on selected matched starts where only GR00T succeeds. Two selected familiar LIBERO goals show opposing advantages: Qwen leads on wine-to-rack, while GR00T leads or ties on cheese-to-bowl. Both systems respond to familiar goal changes. Validation flow loss does not reliably rank development rollout success in these runs. The study evaluates frozen features through downstream control; the behavioral and prediction probes identify where the resulting policies differ, without establishing which representation changes cause those differences.**

***Index Terms*—vision–language–action, robot learning, frozen representations, evaluation, imitation learning**

I. Introduction

Vision–language–action policies connect visual and language representations to robot control [1], [2], [3]. When training a new action head, does a robot-adapted vision–language model (VLM) provide more useful frozen features than its stock origin? We compare Qwen3-VL-2B with the vision–language subtree of GR00T N1.7 using the same head design within each benchmark. The released endpoints differ in their training history [4].

We compare three independently trained heads per backbone. ALOHA Transfer Cube tests visually guided bimanual transfer with a fixed instruction [6]; LIBERO-Goal tests instruction-conditioned single-arm goals [7]. We evaluate frozen-feature usefulness through closed-loop success with matched head architectures. Behavioral and prediction probes then locate differences between the resulting policies. The contributions are a three-seed endpoint comparison, an analysis of task stages and familiar-goal responses, and a comparison of validation loss with development success.

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II. Related Work

Integrated VLA policies [1], [2], [3] and component studies such as RoboVLMs [5] motivate our controlled frozen-feature comparison. Hand views can affect manipulation [8]; our LIBERO heads receive agent and wrist views. Critical Interval MSE [9] and LIBERO-PRO [10] motivate examining offline validation and evaluation conditions, respectively.

III. Experimental Design

A. Backbones and action heads

We compare frozen Qwen3-VL-2B and the vision–language subtree of GR00T N1.7. We cache image and text features at language layer 16. For GR00T this is the retained final layer; for Qwen it is an intermediate layer followed by its final RMS normalization. Within each benchmark, processor settings, view order, token padding, and read depth are matched. Masked padding is removed before real tokens are right-padded to a fixed budget.

Each backbone feeds a separately trained token adapter and DiT-style action decoder [11]. The decoder learns flow matching [12] from demonstrated actions a, noise ϵ, and $t \sim \mathcal{U}(0,1)$:

$$\mathcal{L} = \mathbb{E}\left[\left\|v_\theta((1-t)\epsilon + ta, t, c) - (a-\epsilon)\right\|_2^2\right]. \quad (1)$$

Heads have the same design within a benchmark. ALOHA uses 14.0 million trainable parameters and normalized joint targets; LIBERO uses 19.2 million and raw end-effector actions. Both predict 16 actions, execute eight, and use three inference integration steps before replanning. Their decoder widths are 256 and 320, respectively.

B. Tasks and training

ALOHA Transfer Cube uses a top camera, robot state, and a fixed instruction to control two arms. LIBERO-Goal uses agent and wrist cameras, state, and instructions for ten single-arm goals. ALOHA tests visually guided transfer; LIBERO tests responses to familiar goals. These benchmarks differ in more than language or vision alone.

Both backbones use the same episode split within each benchmark. With split seed 42, ALOHA has 45 training and five validation demonstrations; LIBERO has 450 and 50, with ten percent held out per task. ALOHA normalization uses training data only. We independently train heads with seeds 3, 4, and 5 to epoch 250 for ALOHA and epoch 120 for LIBERO. The configuration specifies highest development success with earliest-epoch tie breaking; the final-test records evaluate endpoint checkpoints. Saved anchor probes use 50 ALOHA or 100 LIBERO development episodes per head. Validation flow loss uses fixed RNG seed 4242. Overlapping validation chunks are not independent episodes, and loss magnitudes are not comparable across the two action spaces.

C. Evaluation

Each ALOHA head is tested on 200 held-out resets; each LIBERO head is tested on 40 resets per task (400 total). These resets are separate from development probes. Within a training-seed pair, the backbones receive matched starts and policy-sampling seeds. We report per-seed success and 95% paired-start percentile bootstrap intervals from 20,000 resamples, stratified by task for LIBERO. The intervals are conditional on the trained heads; the three training seeds, not the rollouts, are the independent training replicates.

To examine goal response, we run each policy from 40 matched starts in each of two scenes with the original and another familiar instruction. The 480 swapped episodes are separate diagnostics. For ALOHA action probes, we replay saved paths with reset, state, and reward checks and query both native systems on the same observations with aligned sampling noise. We also continue two selected Qwen timeouts beyond the original limit. A separate LIBERO audit repeats 18 selected starts for both backbones, twice per head/start (72 episodes), including two previously discrepant starts. These diagnostics do not change the locked final tests.

IV. Progression Results

A. Backbone comparison across training seeds

Table I reports the primary held-out progression tests.

GR00T leads in all three ALOHA seed pairs, with a mean advantage of 23.5 percentage points (sample SD 8.0). In LIBERO the means are both 89.8%, and seed differences change direction (Table I).

Table II lists the ten LIBERO tasks for each seed pair.

B. Where do outcomes differ?

GR00T completes more handoffs among episodes reaching score 2 (Fig. 1A). On matched starts where both policies reached score 2 and only GR00T succeeded, Qwen generally reached that stage later (Fig. 1B). Two selected Qwen timeouts succeeded at steps 489 and 465 when continued beyond the 400-step limit; both native systems also succeeded from their reconstructed step-400 states under each of two shared fresh noise seeds.

LIBERO's aggregate tie conceals goal-level differences. In both scene cohorts, Qwen leads on wine-to-rack in all three seeds, whereas GR00T leads or ties on cheese-to-bowl (Fig. 2). These two goals were selected for closer analysis because their task-level patterns differ; each point compares 40 matched starts. In the Task 6 scene with the swapped wine-to-rack instruction, 16 GR00T failures and 4 Qwen failures contacted the rack without satisfying the goal. In the Task 9 scene with the swapped cheese-to-bowl instruction, GR00T succeeded after 96 of 107 contacts versus Qwen's 86 of 116.

C. Instruction response and action clues

Across 480 episodes with a swapped familiar instruction, neither policy reached the original goal first. Both systems can redirect to familiar goals, with both object and destination changed by the request.

On identical initial observations, changing the instruction also changes layer-16 text features and both heads' predicted first motion commands. In selected outcome-discordant paths, same-state queries yield different action chunks near task events, but the magnitude of the difference does not consistently identify the successful system.

On three selected GR00T-only ALOHA starts, both native systems were queried at each source path's last replanning boundary before score 2. The predicted gripper targets differ on some observations but not others (Fig. 3). All 24 source-policy prefixes across the six selected discordant starts reproduced archived actions exactly. On four selected reset images, masking the cube region changed both endpoints' features, adapter contexts, and actions more than masking an equal-area background patch.

D. Offline loss and development success

Validation loss reaches its minimum at epoch 25 for all twelve heads and then rises, while development success often improves (Fig. 4). Retrospectively, development success at the minimum-loss anchor is below the best plotted probe by a median 50 percentage points in ALOHA and 25.5 points in LIBERO. These best probes need not be eligible for final checkpoint selection. Within-head Spearman correlations between validation loss and development success have medians $+0.271$ and $+0.116$. In these runs, validation flow loss alone does not rank checkpoint rollout success.

TABLE I

PRIMARY PROGRESSION FINAL TESTS. RATES ARE PER TRAINED HEAD. Δ IS GR00T MINUS QWEN IN PERCENTAGE POINTS WITH A 95% PAIRED-START BOOTSTRAP INTERVAL CONDITIONAL ON THE TRAINED PAIR. G-ONLY/Q-ONLY COUNTS ARE MATCHED STARTS WHERE ONLY THAT BACKBONE SUCCEEDS.

Benchmark (n/head)	Seed	GR00T	Qwen	Δ (95% interval)	G-only / Q-only
ALOHA (200)	3	49.5%	34.5%	$+15.0$ ($[7.5, 22.5]$)	47 / 17
	4	42.5%	18.0%	$+24.5$ ($[17.0, 32.0]$)	60 / 11
	5	64.5%	33.5%	$+31.0$ ($[22.5, 39.5]$)	78 / 16
LIBERO-Goal (400)	3	89.0%	89.8%	-0.75 ($[-4.5, 3.0]$)	30 / 33
	4	88.8%	91.8%	-3.0 ($[-6.5, 0.5]$)	22 / 34
	5	91.8%	88.0%	$+3.75$ ($[0.0, 7.5]$)	40 / 25

TABLE II

LIBERO-GOAL TASK SUCCESS RATES (%): GR00T/QWEN, WITH 40 MATCHED STARTS PER HEAD. EACH SEED DENOTES A SEPARATELY TRAINED HEAD PAIR.

Task	Instruction	Seed 3	Seed 4	Seed 5
0	open the middle drawer of the cabinet	90.0/100.0	87.5/95.0	100.0/100.0
1	put the bowl on the stove	97.5/100.0	100.0/97.5	97.5/97.5
2	put the wine bottle on top of the cabinet	97.5/100.0	97.5/100.0	100.0/95.0
3	open the top drawer and put the bowl inside	62.5/75.0	70.0/70.0	87.5/92.5
4	put the bowl on top of the cabinet	100.0/100.0	100.0/97.5	100.0/95.0
5	push the plate to the front of the stove	97.5/90.0	67.5/95.0	90.0/65.0
6	**put the cream cheese in the bowl**	**80.0/60.0**	**90.0/77.5**	**82.5/62.5**
7	turn on the stove	97.5/100.0	100.0/97.5	95.0/95.0
8	put the bowl on the plate	90.0/77.5	87.5/87.5	77.5/82.5
9	**put the wine bottle on the rack**	**77.5/95.0**	**87.5/100.0**	**87.5/95.0**

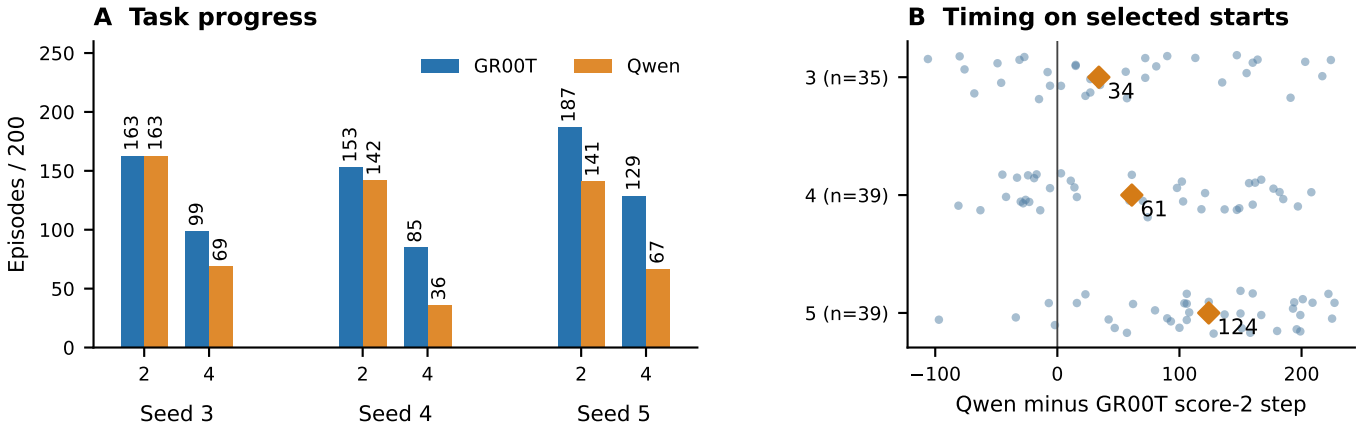


Fig. 1. ALOHA progress and timing. (A) Score 2: right-finger contact with the cube off the table; score 4: successful handoff. (B) Matched starts where both reached score 2 and only GR00T succeeded. Dots are starts; diamonds are median Qwen-minus-GR00T delays. Positive values mean Qwen was later.

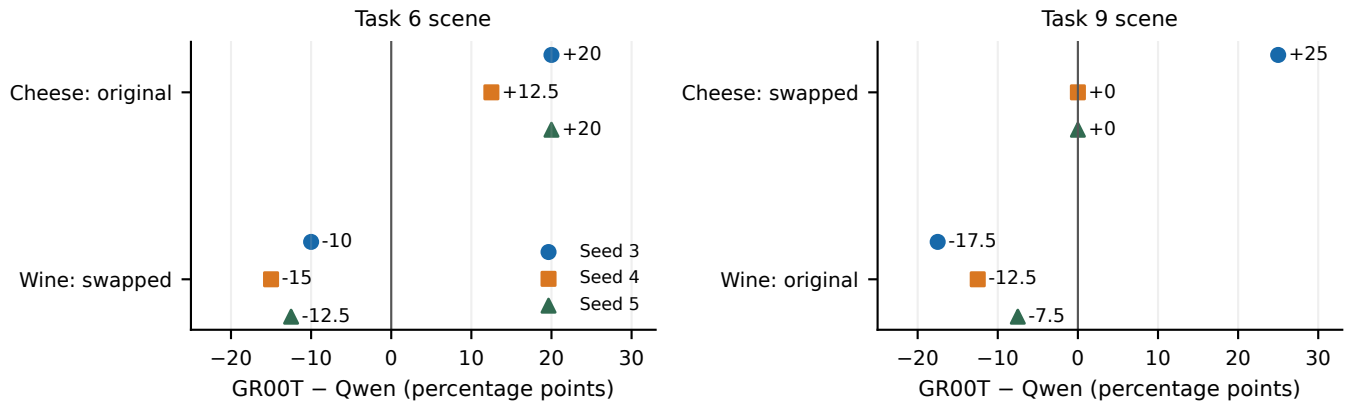


Fig. 2. LIBERO goal comparisons across two scene cohorts. The original task is cheese-to-bowl in Task 6 and wine-to-rack in Task 9; the swapped instruction requests the other goal. Each point is a training-seed pair evaluated on 40 matched starts. Positive differences favor GR00T.

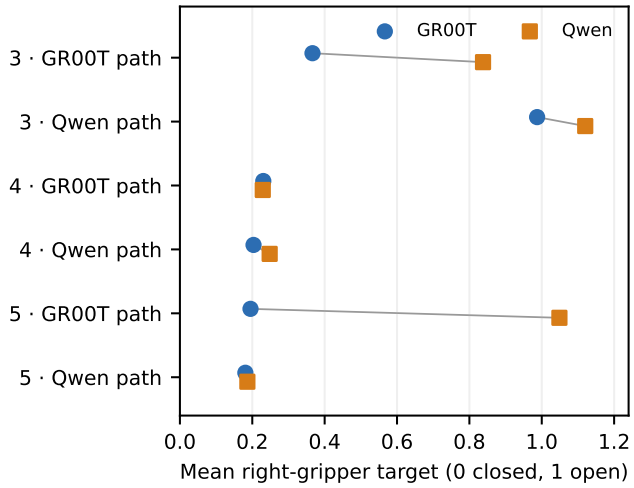


Fig. 3. Predicted right-gripper targets on three selected GR00T-only ALOHA starts. Within each source-path observation, both native systems receive the same image, robot state, and action noise. Markers show mean gripper targets over the next action prefix. Alternative predictions were not executed.

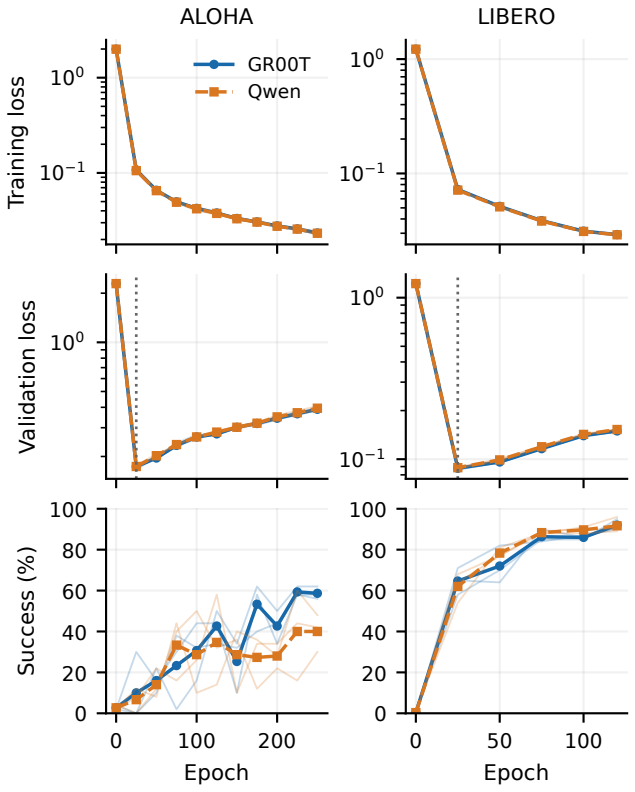


Fig. 4. Training and validation flow loss and development success at checkpoint anchors. Loss axes are logarithmic; faint lines are individual heads and bold lines are three-seed means. The dotted line marks epoch 25. Development probes use 50 ALOHA or 100 LIBERO episodes per head and are separate from held-out final tests.

V. Limitations

Task-level comparisons and outcome-selected probes are descriptive. The timing subsets and two extended episodes do not estimate the prevalence or cause of failures. Alternative action predictions were not executed, and image masking may produce unfamiliar inputs. These probes do not identify which feature changes cause better control. The equal LIBERO means do not establish equivalence, and dependent checkpoint probes do not establish test-set performance or prove overfitting.

The study compares two frozen endpoints from one lineage. The intervening Cosmos-Reason2 model [4] is not tested, so the endpoint contrast cannot be attributed to one pretraining stage. The results concern newly trained heads, not the complete released GR00T policy. ALOHA and LIBERO differ in task, embodiment, camera views, action space, and head size; their contrast cannot isolate a language, vision, or embodiment effect.

The primary comparison uses three independent training seeds per backbone. Per-seed reporting shows the observed training variation; paired-start intervals are conditional on the trained heads. Familiar LIBERO goal swaps do not test unseen language or tasks. Phase measures are simulator predicates. Selected same-state probes use outcome-discordant examples, so they reveal candidate decision points rather than failure prevalence or causal feature effects. The fixed demonstration split, including five ALOHA validation episodes, also limits checkpoint selection coverage.

The LIBERO repeat audit deliberately included two previously discrepant starts. Among 36 selected head/start pairs, both repeats matched the locked outcome in 31, disagreed with each other in four, and both differed from the locked outcome in one; all five were wine-task pairs. All new initial state and both camera hashes matched across repeats, but the locked records lack a separate wrist-image hash. At the first action difference above 10^{-6} in 10 pairs, state and agent-view hashes matched while wrist-view hashes differed. Two fixed-action checks showed occasional one-level 8-bit wrist-render differences with matching simulator states; they do not establish the size or cause of differences in the original repeat episodes. This selected audit is not a benchmark-wide reproducibility estimate.

VI. Conclusion

With matched action-head architectures, frozen GR00T features support higher ALOHA success across all three training seeds, but no aggregate advantage on LIBERO-Goal. GR00T completes more ALOHA transfers among episodes reaching score 2; on selected matched starts where only GR00T succeeds, Qwen reaches that stage later. In LIBERO, the aggregate tie conceals opposing advantages on the two selected familiar goals. Both systems respond to familiar instruction changes. These results evaluate the control usefulness of the frozen features. Timing and action probes locate candidate decision points for further study. Finally, validation flow loss alone did not reliably rank development rollout success in these runs.

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